STAT 330 Exam 1 Study Guide

Modules 1, 2, and 3

Purpose and How to Use This Guide

Exam 1 may assess any topic or skill that appears in the Module 1, 2, or 3 lecture slides, homework, or quizzes. This checklist turns that scope into specific tasks you should be able to complete. It is not a list of the exact exam questions.

Use each box as a self-check. You should be able to explain the idea in words, recognize it in output or a plot, perform the associated calculation, and interpret the result in context when those tasks apply.

What to Expect

- ☐ Questions may use true-false, single-answer, multiple-answer, short-response, calculation, and multipart application formats.
- ☐ You may need to move among a data description, model equation, regression output, diagnostic figures, and a scientific conclusion.
- ☐ Show enough work to make calculations reproducible, and include variables, units, comparison groups, and conditioning statements in interpretations.
- ☐ Treat diagnostic tests and rules of thumb as evidence to weigh. Avoid automatic conclusions such as discarding a model or deleting an observation from one cutoff alone.
- ☐ Distinguish association from causation and distinguish lack of evidence from evidence that an effect is exactly zero.

Module 1 Multiple Regression and Matrix Form

Models and contextual interpretation

- ☐ Identify the response, predictors, observational units, and units of measurement from a study description.
- ☐ Write the theoretical multiple regression model in scalar form and matrix form, including the error distribution and independence assumptions.
- ☐ Distinguish the theoretical model from a fitted equation and distinguish population parameters, estimates, errors, and residuals.
- ☐ Interpret an intercept when it is meaningful and explain when the zero-predictor reference point is outside the useful data range.
- ☐ Interpret a slope as an adjusted association in the conditional mean response, holding all other included predictors constant.
- ☐ Scale a coefficient interpretation to a meaningful change larger than one unit without changing its meaning.

☐ Avoid individual, universal, and causal overclaims when the model supports an average observational association.

Matrix notation and operations

☐ Construct and recognize the response vector Y, design matrix X, coefficient vector $\boldsymbol{\beta}$, and error vector $\boldsymbol{\epsilon}$.

☐ State the dimensions of Y, X, $\boldsymbol{\beta}$, and $\boldsymbol{\epsilon}$ for n observations and p predictors, remembering the intercept column.

☐ Determine whether matrix products are conformable and find the dimensions of valid products.

☐ Use and explain transpose, matrix multiplication, the identity matrix, and a matrix inverse.

☐ Explain why the design matrix contains a column of ones and how one row of X produces one fitted value.

☐ Recognize the matrix products $X^\top X$, $X^\top Y$, $X\widehat{\boldsymbol{\beta}}$, and $Y - X\widehat{\boldsymbol{\beta}}$.

Least squares and fitted values

☐ Explain that least squares minimizes $SSE(\boldsymbol{\beta}) = (Y - X\boldsymbol{\beta})^\top (Y - X\boldsymbol{\beta})$, not the unsquared residual sum.

☐ Derive or explain the normal equations $X^\top X\widehat{\boldsymbol{\beta}} = X^\top Y$ and the estimator $\widehat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top Y$.

☐ State the invertibility condition and explain why perfectly redundant predictor columns prevent a unique coefficient solution.

☐ Calculate fitted values and residuals with $\widehat{Y} = X\widehat{\boldsymbol{\beta}}$ and $e = Y - X\widehat{\boldsymbol{\beta}}$.

☐ Define the hat matrix $H = X(X^\top X)^{-1} X^\top$, use $\widehat{Y} = HY$, and explain why leverage depends on X rather than Y.

☐ Calculate and interpret the unbiased error-variance estimate $\hat{\sigma}^2 = s^2 = e^\top e/(n-p-1)$ and explain the residual degrees of freedom.

Sampling variability and coefficient inference

☐ Distinguish the fixed population vector $\boldsymbol{\beta}$ from the sample-dependent estimator $\widehat{\boldsymbol{\beta}}$.

☐ Explain a coefficient's standard error as repeated-sampling variability and recognize $\widehat{\mathrm{Var}}(\widehat{\boldsymbol{\beta}}) = s^2(X^\top X)^{-1}$.

☐ State coefficient-test hypotheses, calculate a t statistic for a null value that may be nonzero, use $n-p-1$ degrees of freedom, and reach a contextual conclusion.

☐ Use a p-value and significance level correctly. Explain why failing to reject a null hypothesis does not prove the null value is true.

☐ Construct and interpret a coefficient confidence interval, including the repeated-sampling meaning of the confidence level.

☐ Use the agreement between a two-sided level-α test and a matching $(1-\alpha)$ confidence interval.

Prediction and intervals

- ☐ Calculate a point prediction by substituting predictor values into a fitted equation or multiplying a new design row by $\widehat{\boldsymbol{\beta}}$.
- ☐ Distinguish a confidence interval for the conditional mean response from a prediction interval for one new response.
- ☐ Explain why the prediction interval is wider and recognize the additional 1 inside its standard-error formula.
- ☐ Select the appropriate interval from the wording of a scientific question and interpret it using the response, predictor profile, units, and target population.

Module 2 Diagnostics Model Checking and Transformations

LINE assumptions and diagnostic evidence

- ☐ State the LINE assumptions: linearity, independence, normality of errors, and equal error variance.
- ☐ Explain why diagnostics evaluate the adequacy of a model rather than prove that assumptions are exactly true.
- ☐ Assess linearity using residuals-versus-fitted plots and added-variable plots, keeping curvature separate from changing variance.
- ☐ Explain what an added-variable plot isolates and why a raw response-versus-predictor plot can be misleading in multiple regression.
- ☐ Assess independence from study design, sampling, time order, clusters, and repeated measurements rather than from an ordinary residual plot alone.
- ☐ Assess normality using a residual histogram, Q-Q plot, and Shapiro-Wilk test. State the test hypotheses and weigh statistical evidence against practical severity.
- ☐ Assess equal variance using residuals-versus-fitted and scale-location plots together with the Breusch-Pagan test. State the test hypotheses.
- ☐ Explain the likely consequences of each violated assumption for coefficient estimates, standard errors, tests, confidence intervals, and prediction intervals.

Measures of fit and overall inference

- ☐ Calculate and connect SSE, SSR, and SST, including $SST = SSR + SSE$ for a model with an intercept.
- ☐ Calculate MSE and RMSE, state their units, and distinguish a training error from a held-out prediction error.
- ☐ Calculate and interpret R^2 as the proportion of observed response variation explained by the predictors together.
- ☐ Explain why ordinary R^2 cannot decrease when predictors are added and why adjusted R^2 can decrease.
- ☐ State the overall F-test hypotheses, identify its degrees of freedom, and interpret a significant or nonsignificant result.

☐ Explain why a significant overall F-test does not identify which predictors matter and can coexist with nonsignificant individual tests.

Leverage residual size and influence

☐ Distinguish leverage, a standardized residual, and influence. Explain why none of the three is interchangeable with the others.

☐ Calculate or identify leverage from the diagonal of the hat matrix and explain what unusual predictor values mean.

☐ Calculate Cook's distance from a standardized residual, leverage, and the number of coefficients.

☐ Apply rules of thumb such as $4/n$ as flags for investigation, not automatic deletion rules.

☐ Use a residuals-versus-leverage display to identify observations that deserve follow-up.

☐ Describe appropriate follow-up: check data quality and context, assess population membership, and compare analyses with and without the observation.

Transformations

☐ Explain why transforming a predictor, response, or both can address curvature, changing spread, or nonnormal errors.

☐ Recognize common log, square-root, and reciprocal transformations and describe how they stretch or compress a scale.

☐ Refit a transformed model and recheck all relevant diagnostics instead of assuming the transformation solved every problem.

☐ Interpret coefficients in level-level, level-log, log-level, and log-log models.

☐ Use exact expressions involving e^{β} and appropriate small-coefficient percentage approximations.

☐ Explain why exponentiating a fitted log response estimates a conditional median unless a separate mean-bias correction is used.

☐ Back-transform point predictions and interval endpoints correctly.

☐ Compare predictive errors on a common response scale and avoid comparing R^2 values across different response scales without qualification.

Module 3 Multicollinearity Model Selection and Shrinkage

Multicollinearity

☐ Explain how correlated or redundant predictors affect coefficient stability, signs, standard errors, individual tests, and interpretation.

☐ Explain why multicollinearity can produce a significant overall model test while individual slopes are uncertain.

☐ Use a correlation display as an initial diagnostic while recognizing that pairwise correlations do not capture every multivariable dependency.

- ☐ Calculate $\text{VIF}_j = 1/(1 - R_j^2)$ from an auxiliary regression and calculate the standard-error inflation factor $\sqrt{\text{VIF}_j}$.
- ☐ Interpret VIF as coefficient-variance inflation and avoid treating a threshold as an automatic variable-removal rule.
- ☐ Connect perfect multicollinearity to noninvertibility and imperfect multicollinearity to a nearly unstable coefficient solution.

Search algorithms and selection metrics

- ☐ Distinguish the set of candidate models, the algorithm used to search them, and the metric used to score them.
- ☐ Describe best-subsets search and explain why it is exhaustive over the permitted predictor combinations.
- ☐ Describe forward selection, backward elimination, and stepwise search as greedy path-based algorithms.
- ☐ Explain why a greedy search can miss the globally best model for a chosen size or metric.
- ☐ Distinguish ordinary R^2, adjusted R^2, AIC, and BIC, including the direction that indicates a preferred model.
- ☐ Explain why AIC, BIC, and adjusted R^2 can select different models because they penalize complexity differently.
- ☐ Explain why repeated individual p-value screening is a poor model-search strategy.
- ☐ Separate an in-sample selection criterion from an out-of-sample prediction criterion.

Training test splits and cross-validation

- ☐ Explain why evaluating a model on observations used to fit it gives an optimistic performance estimate.
- ☐ Identify the roles of training data and test data and keep the test responses out of model fitting.
- ☐ Put repeated random-split validation steps in order: initialize storage, split, fit on training data, predict test data, calculate test RMSE, repeat, then summarize.
- ☐ Explain why all candidate models should use the same splits and how paired RMSE differences improve comparisons.
- ☐ Select the smaller mean cross-validated RMSE while considering split-to-split variability and practical importance.
- ☐ Explain how a model can have the best training fit but weaker held-out prediction because of overfitting or high variance.
- ☐ Distinguish repeated train-test splitting from a single split and from other resampling designs.

Bias variance ridge and LASSO

- ☐ Explain the bias-variance tradeoff and why adding bias can improve prediction by reducing variance.

- ☐ Write or recognize the OLS, ridge, and LASSO objectives and identify the L_2 and L_1 penalties.
- ☐ Explain the role of λ: larger values impose stronger shrinkage, while $\lambda = 0$ returns the unpenalized objective.
- ☐ Explain why predictors are standardized before applying a common shrinkage penalty.
- ☐ Read coefficient-path and cross-validation plots as λ changes.
- ☐ Distinguish $\lambda_{\min}$ from λ_{1se} and explain the prediction-versus-parsimony tradeoff.
- ☐ Explain why ridge generally retains all predictors while LASSO can set coefficients exactly to zero.
- ☐ Choose ridge when many correlated predictors may share signal and choose LASSO when a sparse, interpretable model is important.
- ☐ Explain how shrinkage can be used when the number of predictors exceeds the number of observations.
- ☐ Avoid claiming that ridge or LASSO automatically provides classical OLS confidence intervals or hypothesis tests.

Complete modeling workflow

- ☐ Move from a scientific question through exploration, model fitting, diagnostics, model comparison, final model checking, and communication.
- ☐ Recheck LINE assumptions, influence, and multicollinearity for a selected final linear model.
- ☐ Choose a final model according to the stated goal, such as explanation, parsimony, or prediction, rather than declaring one method universally best.
- ☐ Communicate conclusions to a nontechnical audience with appropriate uncertainty and without causal overstatement.

Core Formula Checklist

You should recognize each formula, know what its pieces mean, and know when it is appropriate.

$$Y = X\boldsymbol{\beta} + \boldsymbol{\epsilon}, \qquad \widehat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top Y,$$

$$\widehat{Y} = X\widehat{\boldsymbol{\beta}} = HY, \qquad H = X(X^\top X)^{-1} X^\top,$$

$$e = Y - \widehat{Y}, \qquad s^2 = \frac{e^\top e}{n - p - 1},$$

$$\widehat{\mathrm{Var}}(\widehat{\boldsymbol{\beta}}) = s^2 (X^\top X)^{-1}, \qquad t = \frac{\widehat{\beta}_j - \beta_{j,0}}{SE(\widehat{\beta}_j)},$$

$$R^2 = 1 - \frac{SSE}{SST}, \qquad \mathrm{VIF}_j = \frac{1}{1 - R_j^2},$$

$$D_i = \frac{r_i^2}{p+1} \left(\frac{h_{ii}}{1 - h_{ii}} \right), \qquad \mathrm{RMSE} = \sqrt{\frac{1}{m} \sum_{i=1}^{m} (y_i - \widehat{y}_i)^2}.$$

$$\text{Ridge:} \quad \min_{\beta_0,\boldsymbol{\beta}} \left\{ \sum_i (y_i - \beta_0 - x_i^\top \boldsymbol{\beta})^2 + \lambda \sum_j \beta_j^2 \right\},$$

$$\text{LASSO:} \quad \min_{\beta_0,\boldsymbol{\beta}} \left\{ \sum_i (y_i - \beta_0 - x_i^\top \boldsymbol{\beta})^2 + \lambda \sum_j |\beta_j| \right\}.$$